SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	G. Himabindu	**Reg. No.:**	192511377
Section / Batch:	B.Tech AIML	**Date:**	15-07-26.

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
	GRAND TOTAL:		**100 Marks**

SECTION A – SHORT ANSWER TEST

1.Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.
(a) Enumerate three classes of schemas that are popularly used for modelling data warehouses.
(b) Draw a schema diagram for the above data warehouse using one of the schema.
(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).
(5 Marks)

2.Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)
(a) Draw a snowflake schema diagram for the data warehouse.
(b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big University student.
(c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?

3.Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)
Assume that there are no concept hierarchies associated with the dimensions.
(a) What is the maximum number of cells possible in the base cuboid?
(b) What is the minimum number of cells possible in the base cuboid?
(c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the data cube, C?
(d) What is the minimum number of cells possible in the data cube, C?

5. 1. University Academic Performance Data Warehouse
A university maintains a data warehouse to analyse student academic performance. The warehouse consists of the following dimensions:

- Student: Department, Year of Study, Gender
- Course: Program, Semester, Subject
- Faculty: Department, Designation
- Time: Year, Semester, Month

Measures:

- Total Marks
- Attendance Percentage

At the lowest conceptual level, Total Marks stores the marks obtained by each student in a subject. At higher conceptual levels, it stores the average marks.
Questions
(a) Draw a star schema for the university academic performance data warehouse.
(b) Starting from the base cuboid [Student, Course, Faculty, Time], identify the OLAP operations required to display the average marks department-wise for each academic year.
(c) If Student has 4 levels, Course has 3 levels, Faculty has 3 levels, and Time has 5 levels, calculate the total number of cuboids.

Code of Conduct
I G.Hemabindu (A25511377) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: G.Hima

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

G. Himabindu
192511377.

CO1 Assessment Tool-1

Hospital Data Warehouse.

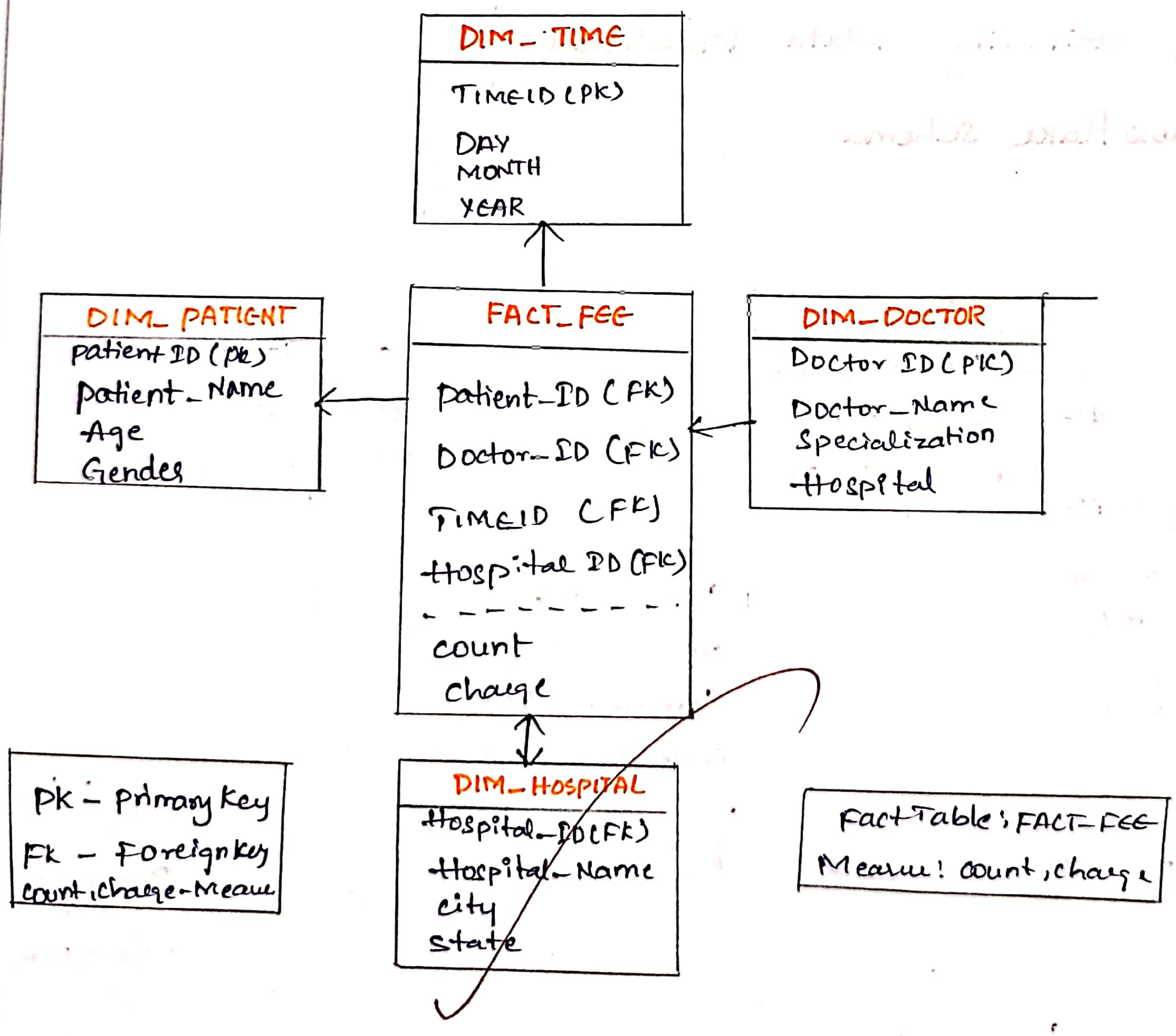
a) Three popular schemas:-

1) Star Schema

2) Snowflake schema

3) Fact constellation (Galaxy) schema

b) STAR SCHEMA – CITY Hospital Data warehouse.

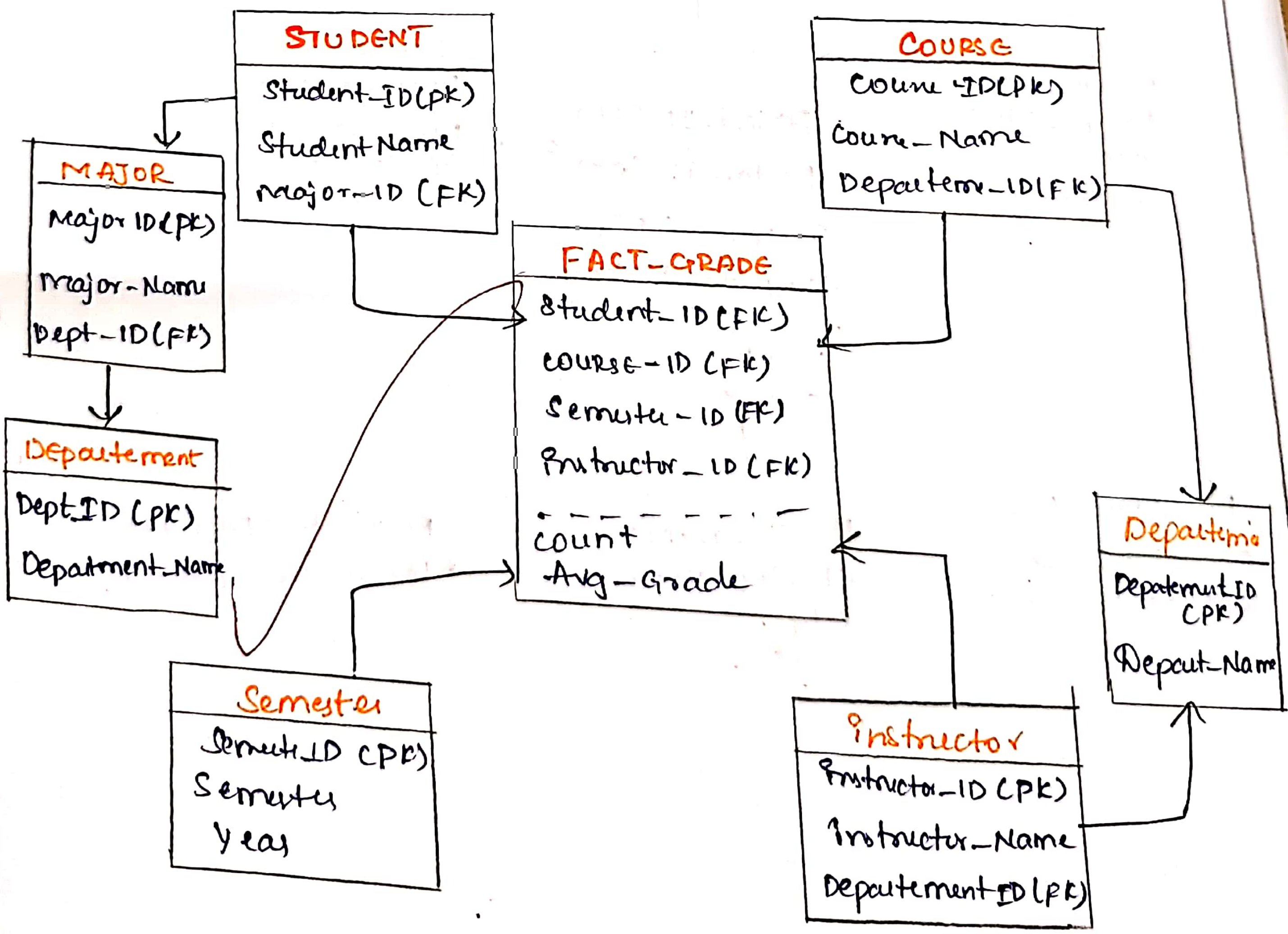


c) SQL Query:-

SELECT day, Month, year, doctor, hospital, patient.
SUM(count) As total_count,
SUM (charge) As total_charge
FROM fee
GROUP BY day, month, year, doctor,
hospital, Patient ;

② Big university Data Warehouse

a) Snowflake schema



PK - Primary Key   FK - Foreign key.

operations :-

rollup student – university

slice course = CS

rollup semester → Academic year

Drill - down if required to individual students.

To calculate Numbers of cuboids

Formula:-

$n_1 \times n_2 \times n_3 \times n_4$

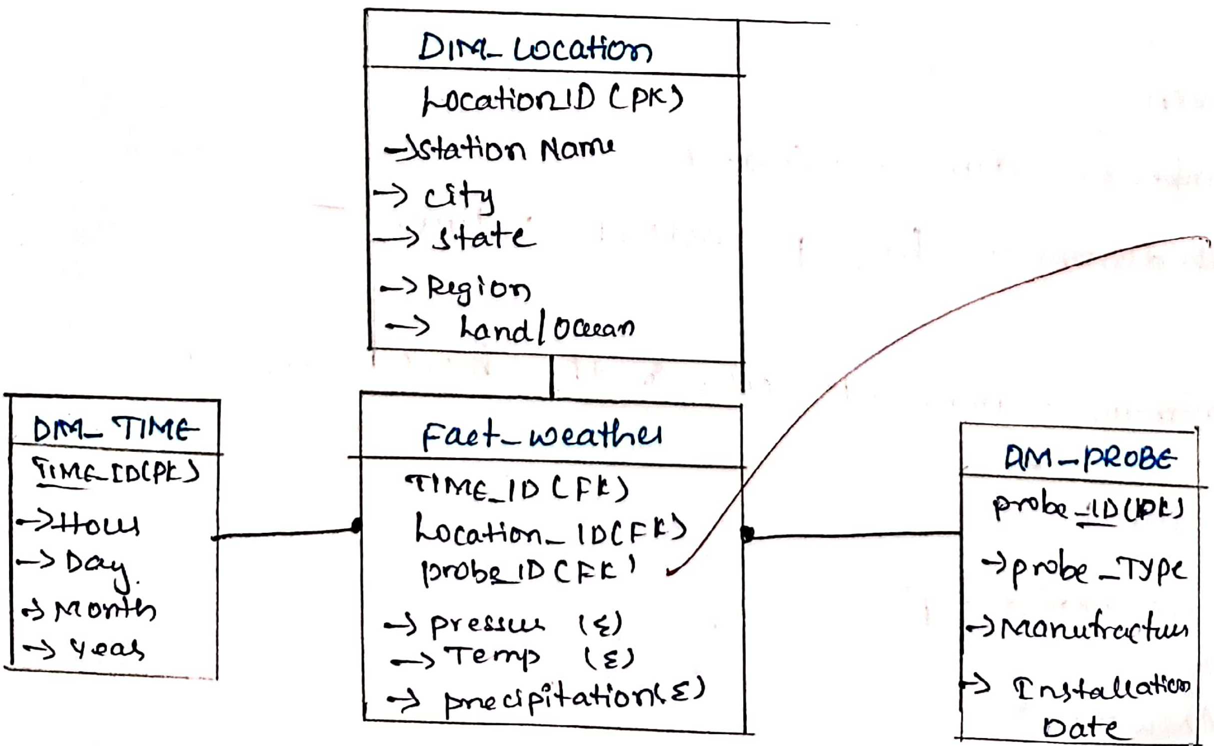
Each dimension has 5 levels

$5^4 = 625$

∴ Answer = 625 cuboids.

3)

4) Star schema for Regional weather Bureau Datawarehouse



PK → primary key   FK → Foriegn key   Σ - Measure.

Dimensions:-

- Time :- Hour, Day, Month, year
- Location :- Station, City, State, Region, Land, ocean
- probe :- probe ID, Type, Manufacturer

Measures:-

- Temperature
- pressure
- precipitation

Advantages:-

- Fast OLAP analysis
- Easy multidimensional reporting
- Supports trend analysis over 10+ years
- Efficient roll-up and drill-down operations.

4) Given

Number of dimensions $(n) = 6$

Each dimension has p distinct values.

a) Maximum number of cells in the base cuboid

Formula:

Base cuboid $= p^6$

$\therefore$ Answer $= p^6$.

Minimum number of cells in the base cuboid.

Since every combination must exist,

Minimum $= p^6$.

c) Maximum total number of cells in the data cube (including aggregate cells)

Formula:-

$(p+1)^6$

$\therefore$ Answer : $(p+1)^6$

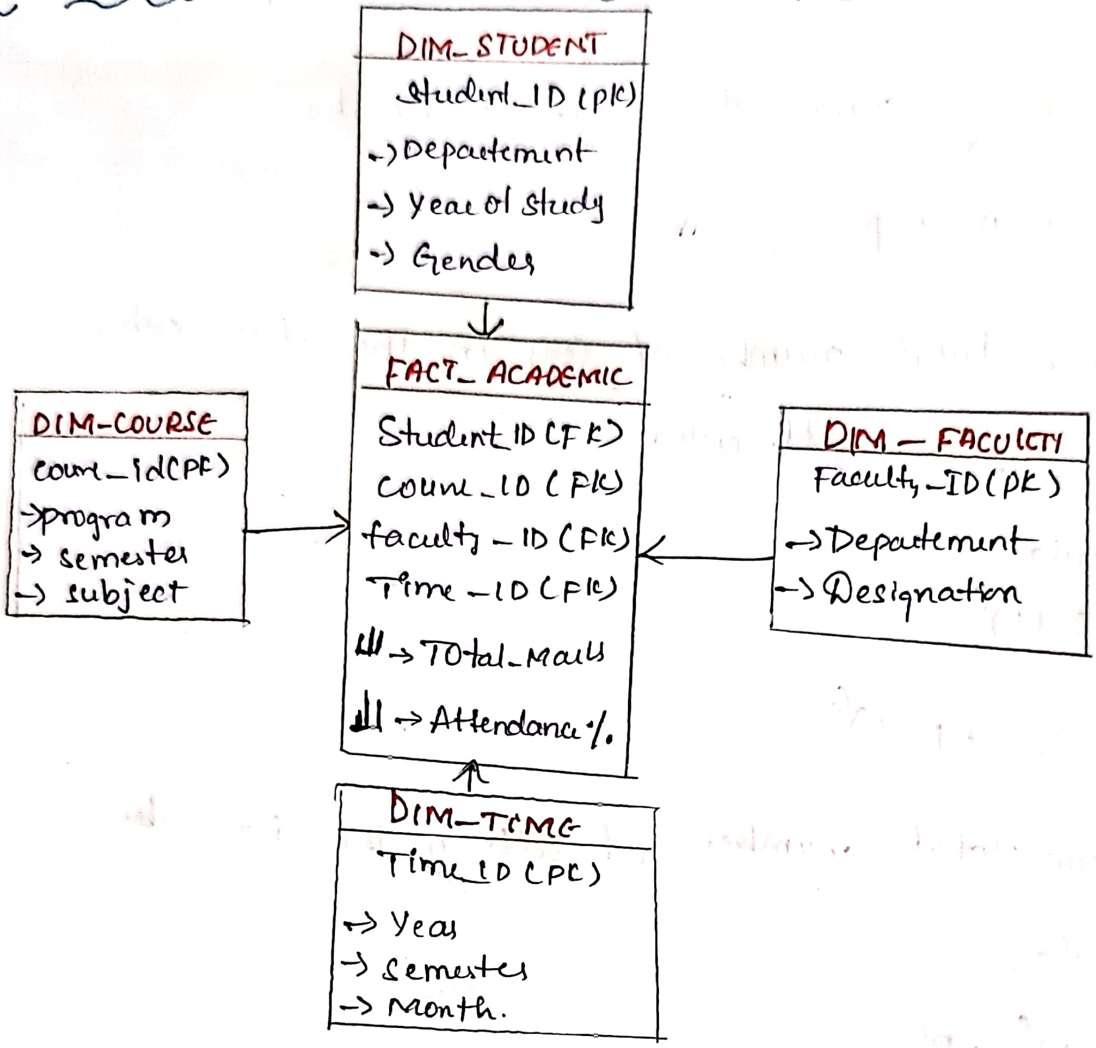
d) Minimum total number of cells in the data cube

Formula:

$(p+1)^6 - p^6$

$\therefore$ Answer $= (p+1)^6 - p^6$

5) a) STAR SCHEMA for University Academic performance datawarehouse



b) OLAP Operations :-

1) Rollup :- → Student = Departement
→ Time = Academic year

2) slice : Select Average Marks measure

3) Dice : Filter a specific faculty or course if required

4) Aggregate : calculate AVG (Total Marks) for each Departement and Academic Year.

c) Total No. of cuboids

Total = product of hierachy levels
= Student × course × faculty × Time
= 4 × 3 × 3 × 5
= 180

∴ Answer = 180 cuboids.